Rex Chong (He/Him) **Electrical Engineering**

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TECHNICAL SKILLS

Programming
- C / C++
- Java / JavaFX
- RISC-V / 8051 Assembly
- SystemVerilog

Embedded & Hardware:
- Autodesk Fusion
- MATLAB
- QuestaSim
- Quartus

Software & Tools:
- FSM design
- Arduino
- Reflow Soldering
- Embedded Systems

EDUCATION

University of British Columbia **April 2029**
Bachelor of Applied Science - Electrical Engineering **GPA: 89.2%**

WORK EXPERIENCE

Math Tutor Network **September 2025 - Current**
Junior Academic Tutor

- Responsible for providing pre-calculus tutoring for 4 high-school students on a weekly basis
- Independently handled lesson scheduling, planning, and payment collection with clients
- Often receive returning offers from clients after each semester

TECHNICAL PROJECTS

Magnetic Field Line-Following Robot (*UBC ELEC 291/292 Project)* **Jan 2025 – Feb 2025**

- Designed and implemented a remote-control autonomous robot using PIC32 (controller) and STM32 (robot) communicating via IR and Bluetooth.
- Developed magnetic field sensing and path-following algorithm using inductive pickup coils and intersection-based navigation logic
- Implemented manual and autonomous control modes, including joystick-based PWM motor control and pre-programmed/custom paths
- Built two-way Bluetooth interface to stream real-time robot telemetry (path progress, battery status) and upload custom paths via CSV
- Integrated object detection and automated claw mechanism using IR sensors for pickup/drop-off tasks

Reflow Oven Design, UBC (Team Course Project for ELEC291) ***Feb. 2026***

- Converted a consumer toaster oven into a programmable reflow soldering system
- Implemented a finite state machine in 8051 assembly to execute customizable reflow profiles (preheat, soak, reflow, cooling)
- Designed thermocouple-based temperature acquisition system with cold junction compensation and ADC for real-time closed-loop control
- Implemented PWM-based time-proportional control of heating elements via solid-state relay; achieved ±1 °C temperature regulation
- Developed user interface with LCD display, speaker alerts, and fault detection logic
- Integrated IR sensor and motor for automated actuation and system sequencing
- Utilized hardware timers and interrupt service routines to ensure deterministic control timing

Amortization program, Personal Project **June 2024 – Sept. 2024**

- *Designed and developed a Windows desktop application for mortgage amortization analysis using JavaFX*
- *Developed responsive GUI using JavaFX and CSS for structured user input and data visualization*
- *Packaged application with a functional Windows installer for end-user deployment*
- *Integrated print functionality for formatted amortization reports*

VOLUNTEER EXPERIENCE

Hope Lutheran Church, Coquitlam, BC **Feb. 2022 – May 2025**
Youth Leader

- Developed strong organization skills by designing and organizing events for 20-40 children.
- Gained valuable teamwork and leadership experience through working with a group of 5-10 leaders to host events.
- Improved problem-solving abilities with flexible thinking due to adaptation to unexpected situations.

AWARDS

- Governor General's Bronze Medallion 2024
- BC Achievement Scholarship 2024
- Dean's List 2025

INTERESTS & ACTIVITIES

- **Percussion** – RCM level 10 percussionist
- **Martial Arts** – Black belt in Hapkido martial arts
- **Drone flying** – Possesses a drone license certification for advanced operations